

SEOKHYEON LEE

☎ +82 10 2315 3965 ✉ winno94@snu.ac.kr 🏠 Oct. 10, 2005

Introduction

Third-year undergraduate in Computer Science and Engineering at Seoul National University. Passionate about physical AI, leveraging computer vision to improve human quality of life, and building efficient large-scale language models. Eager to advance AI systems that are both practically impactful and computationally sustainable.

Education

Seoul National University **Mar. 2024 – Present**
B.S. Department of Computer Science and Engineering

Relevant Coursework

Completed	In Progress	Planned
• Logic Design • Computer Programming • Discrete Mathematics • Algorithms • AI and Law	• Data Structure • Computer Architecture • Advanced Algorithms (graduate course)	• Artificial Intelligence • Deep Learning • Computer Vision • Machine Learning

Academic Projects

64-Bit RISC-V Pipelined CPU & Cache Architecture | *Verilog, Vivado, C* **Spring 2026**

- Designed and implemented a 64-bit RISC-V CPU from single-cycle to a 5-stage pipelined architecture in Verilog; resolved data and control hazards via a custom forwarding unit and static branch predictor relocated to the ID stage.
- Developed a 2-level inclusive cache simulator in C, evaluating LRU and FIFO eviction policies on realistic memory access traces.

Matrix Multiplication Optimization on RISC-V | *C, RISC-V Assembly* **Spring 2026**

- Optimized $1024 \times 1024 \times 4096$ matrix multiplication on a RISC-V 64GC processor, applying cache blocking, loop unrolling, loop reordering, and RISC-V SIMD inline assembly to maximize computational throughput.

SportsCare | *Java, Android Studio, Firebase, Python, MediaPipe* **May 2026 – Present**

- Developing an AI-powered Android app that analyzes user-uploaded sports videos to provide posture correction feedback as a personal training assistant.
- Implemented Firebase Authentication, gallery video upload to Firebase Storage, and real-time Firestore listener for receiving AI analysis results.

MCMC Estimation of Average Independent Set Size | *Python* **May 2026 – Present**

- Applying Markov Chain Monte Carlo (Glauber dynamics) to estimate average independent set sizes under the hard-core model, as a graduate-level Advanced Algorithms final project.

Technical Skills

Languages: Python, Java, C++, Verilog, RISC-V Assembly
Developer Tools: VS Code, Android Studio, Vivado, Docker, Google Colab

Extracurricular

Club Soccer301 **Fall 2024 – Present**
Secretary *Seoul National University*

- Organize and manage club activities, including scheduling meetings and coordinating events.

Campus Mentoring Program **Sep. 2025 – Dec. 2025**
Mentor *Seoul National University*

- Mentored 2 incoming students on academic planning and campus adjustment through regular one-on-one sessions.